

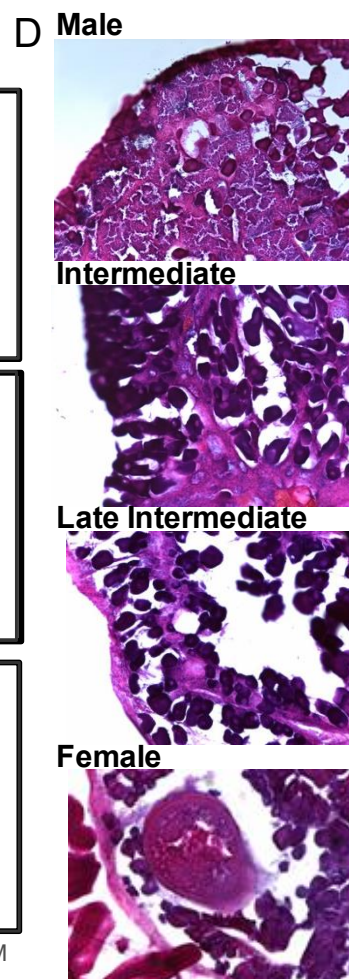
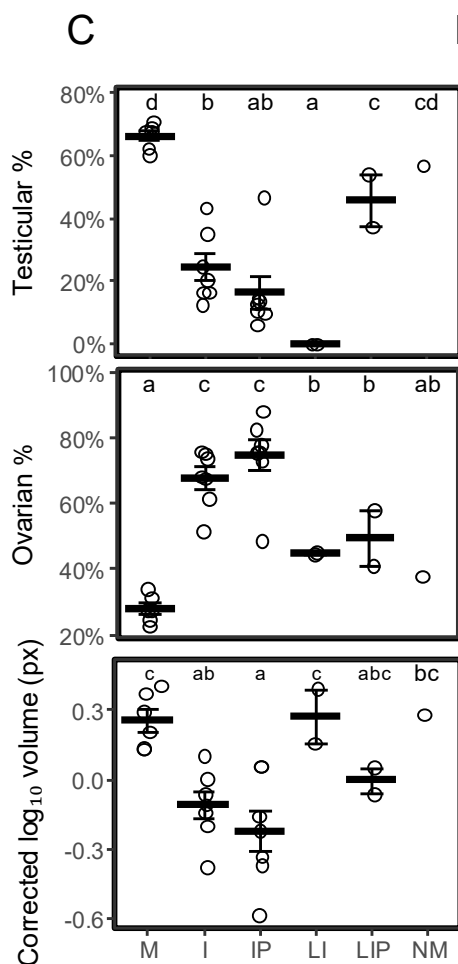
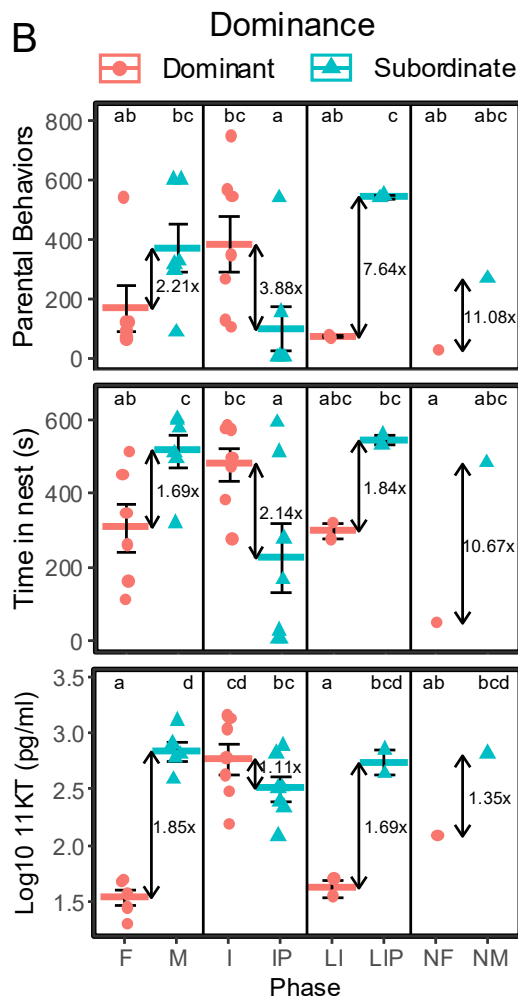
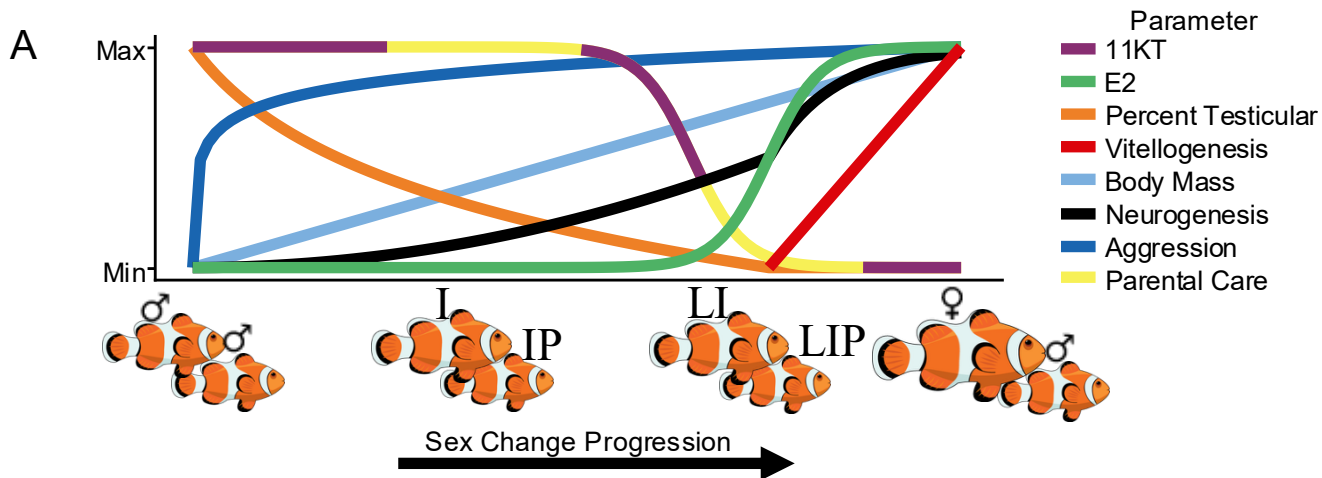


Fig.1. Male, intermediate, and female anemonefish display distinct physiology and behavior. (A) Schematic outlining the behavioral and physiological changes that occur during protandrous sex change in anemonefish.

(B) Top panel: Parental behaviors observed during a parental care assay. Middle panel: Time spent in the nest during parental care assay; a proxy for parental care. Bottom Panel: Log10 transformed serum 11-ketotestosterone.

(C) Top panel: Percent of the gonad made of testicular tissue. Middle panel: Percent of the gonad made of ovarian tissue. Bottom panel: Gonadal volume corrected for the individual's length (methods).

(D) Example hematoxylin and eosin stains of gonads from the major phases of sex change. In B, plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color. Arrows indicate fold-difference between the two groups. In B, C, crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different.

Fig.2. Multiomics capture high quality transcriptomes, epigenomes, and spatial transcriptomes,

- (A) UMAP embeddings of our sn-Multiome data with each cluster numbered.
- (B) Dotplot showing canonical and data-driven markers for each of our multiome clusters.
- (C) Heatmap showing the scaled proportion of each neuroanatomical region made of each multiome cluster (see supplement xx for abbreviations).
- (D) Example projections of multiome clusters onto representative brain sections. The dashed line indicates approximately where we dissected brains for sn-multiomics analysis. Below each section is a sketch of an *A. ocellaris* brain with a line indicating the approximate sectioning plane for the given section.

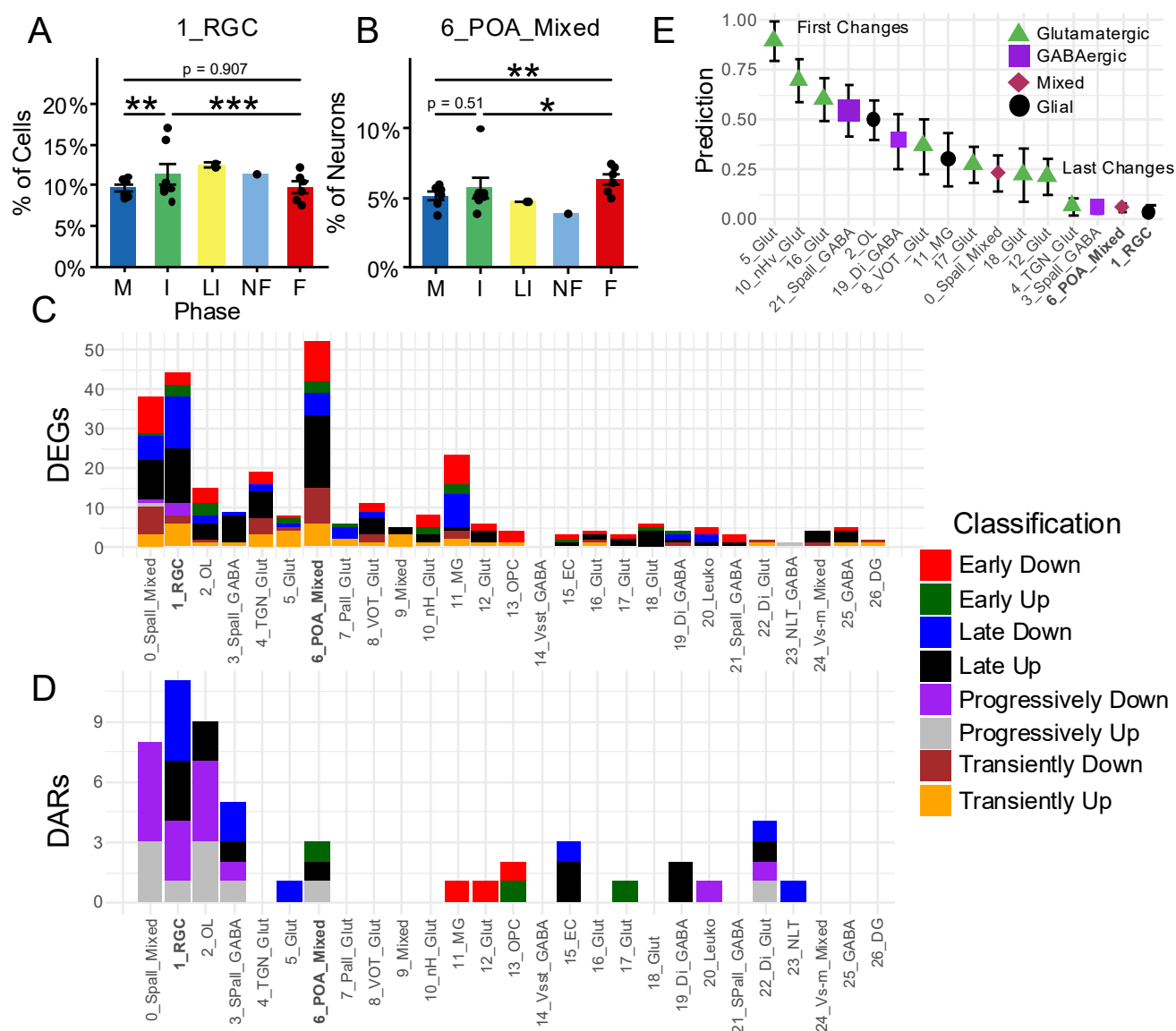


Fig.3. Cells change composition, transcriptome, and epigenome asynchronously during sex change.

(A) The percent of all cells belonging to 1_RGC.

(B) The percent of all neurons belonging to 6_POA_Mixed.

(C) The number of differentially expressed genes (DEGs) by multiome cluster.

(D) The number of differentially accessible regions (DARs) by multiome cluster.

(E) Machine learning predictions for the mean sex of intermediates based on their expression of DEGs and DARs by multiome cluster; a proxy for the temporal order of sex change Shape and color represents broad cell type.

In C, D, bars are colored based on our classification for the manner in which they change sex (see methods). In A, B, points indicate the proportion for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

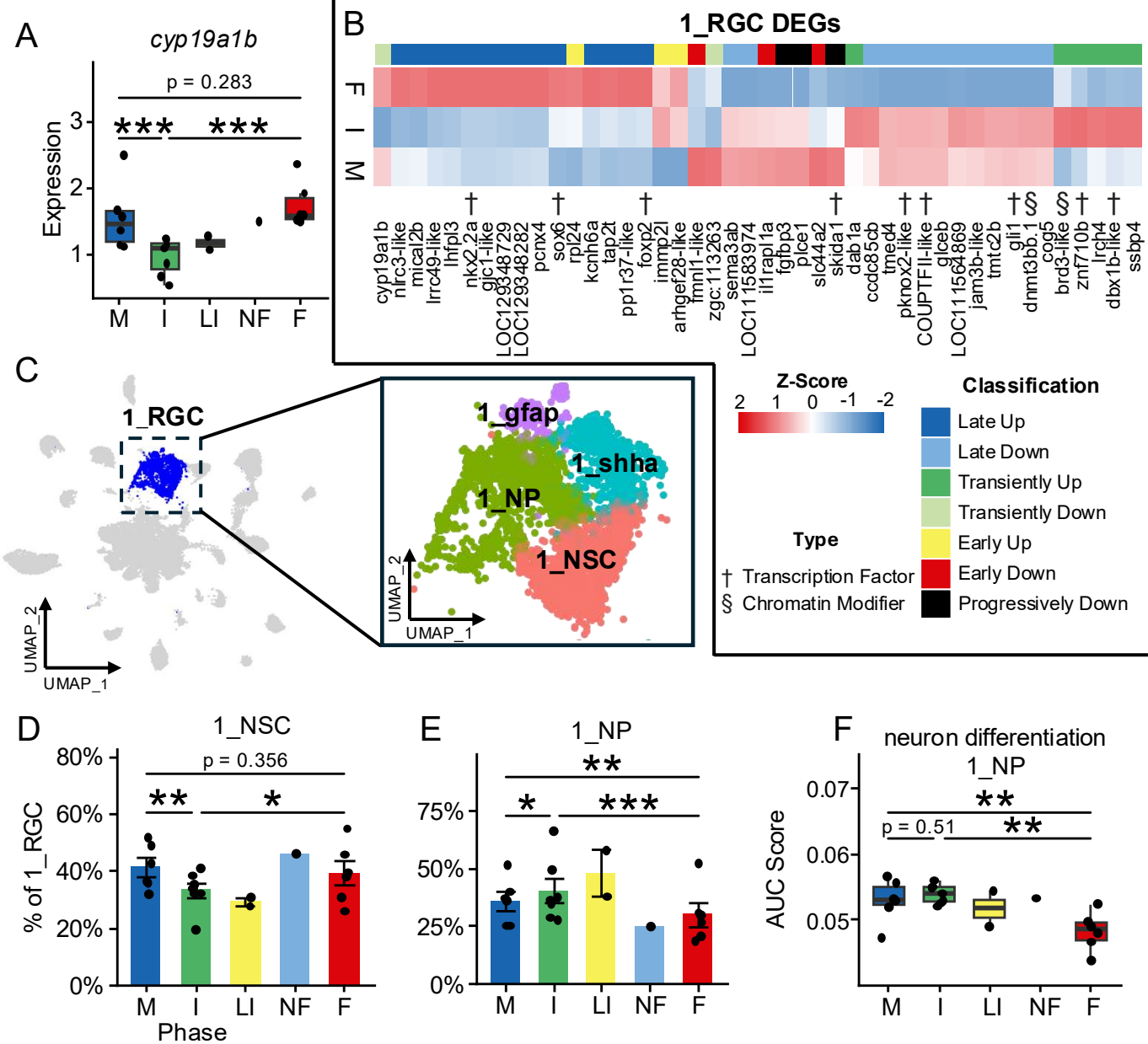


Fig.4. Radial glial cells undergo changes in transcriptome and subtype composition over the course of sex change.

(A) Brain aromatase (*cyp19a1b*) expression in 1_RGC.

(B) A heatmap showing all DEGs by phase in 1_RGC. Cells are colored by z-score of the mean expression for each phase. The top bar indicates the classification for how each gene changes sex, and the symbols below the matrix indicate what kind of gene each DEG is.

(C) UMAP showing subclustering of 1_RGC.

(D) Percent of 1_RGC composed of 1_NSC.

(E) Percent of 1_RGC composed of 1_NP.

(F) AUCell score of a gene expression module composed of genes labeled with the GO term “neuron differentiation” in 1_NP.

In A, D-F, points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

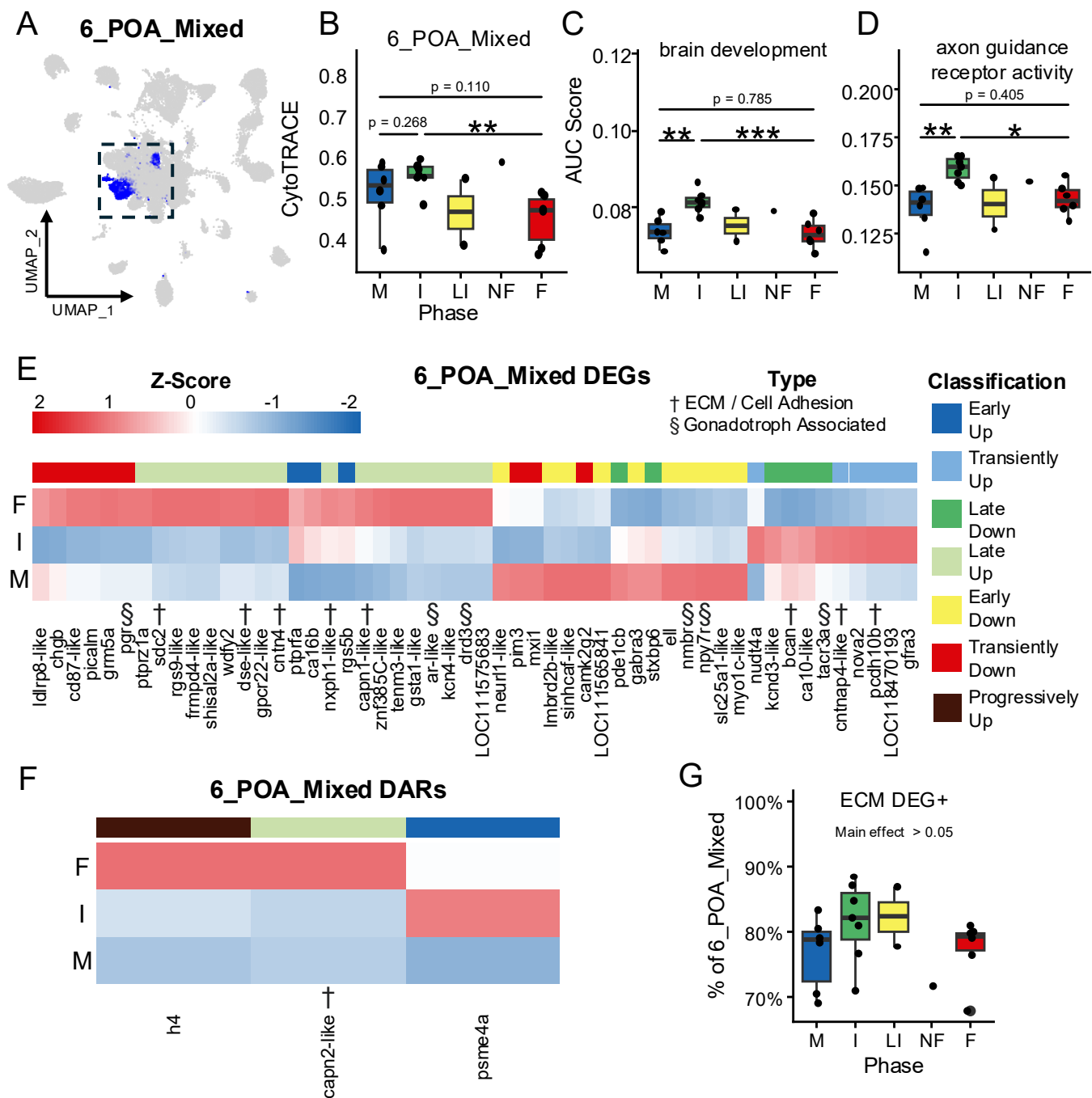


Fig.5. 6_POA_Mixed undergoes changes in transcriptome and epigenome consistent with neurogenesis.

(A) UMAP projection with 6_POA_Mixed highlighted.

(B) CytoTRACE score for 6_POA_Mixed as a whole.

(C) AUCell score for a gene expression module made of genes annotated with the GO term “brain development”

(D) AUCell score for a gene expression module made of genes annotated with the GO term “axon guidance receptor activity”.

(E) DEGs in 6_POA_Mixed. In B, C, cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed.

(F) DARs in 6_POA_mixed..

(G)Prortion of cells in 6_POA_Mixed that are ECM DEG+

In B-D, G, points indicate the mean for each individual. Asterisks: “*” : $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

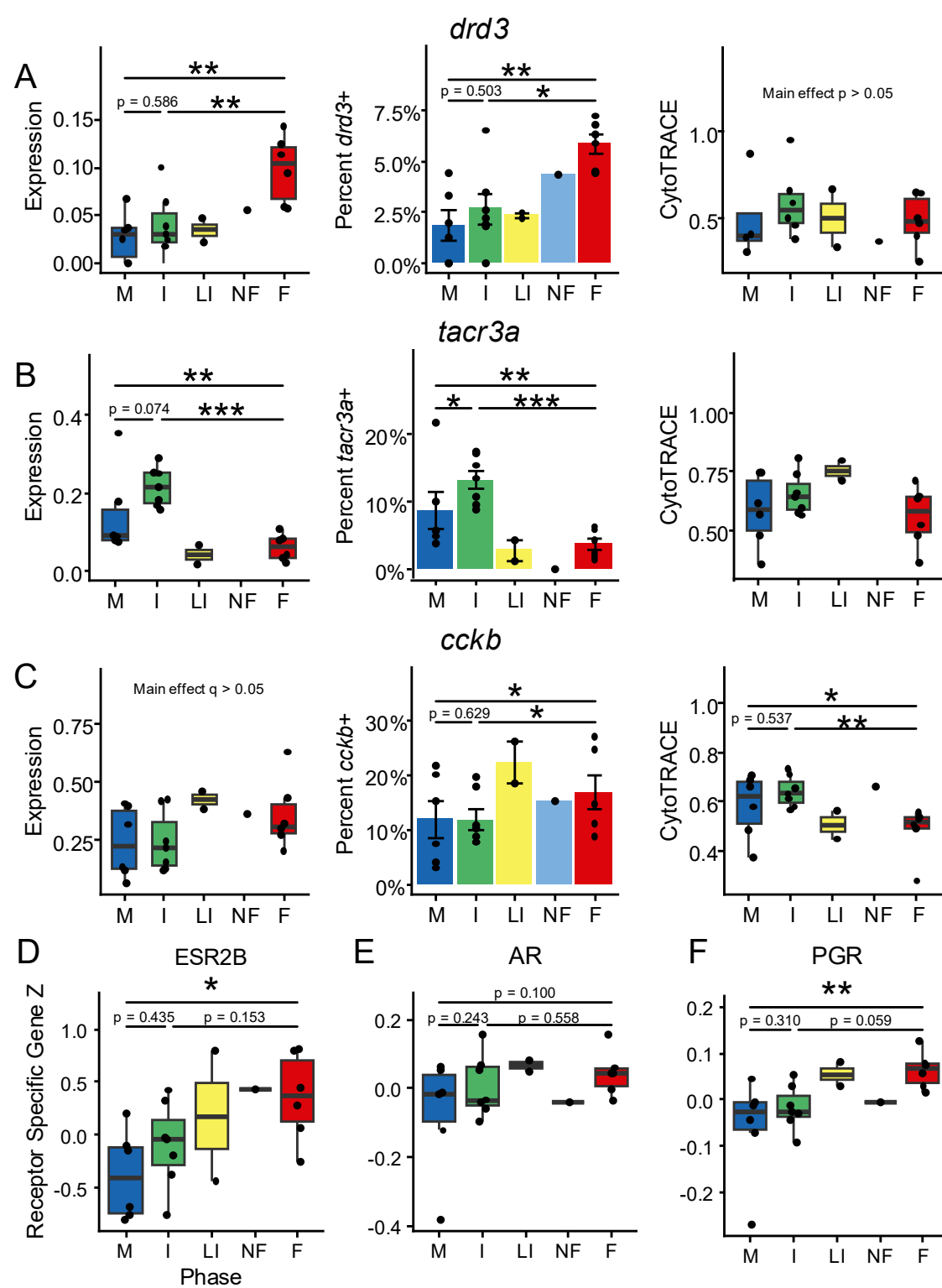


Fig.6. Changes to 6_POA_Mixed Drive Differential Regulation of Gonadotrophs.

- (A) Left panel expression of *drd3* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *drd3* by phase. Right panel: CyoTRACE score for *drd3*+ cells by phase.
- (B) Left panel expression of *tacr3a* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *tacr3a* by phase. Right panel: CyoTRACE score for *tacr3a*+ cells by phase.
- (C) Left panel expression of *cckb* by phase. Middle panel: Percent of 6_POA_Mixed cells expressing *cckb* by phase. Right panel: CyoTRACE score for *cckb* + cells by phase.
- (D) xx justin can you describe what these are
- (E) xx justin can you describe what these are
- (F) xx justin can you describe what these are

Points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$. “Main effect Main effect $p > 0.05$ ” indicates type III tests were not significant so pairwise comparisons were not computed.

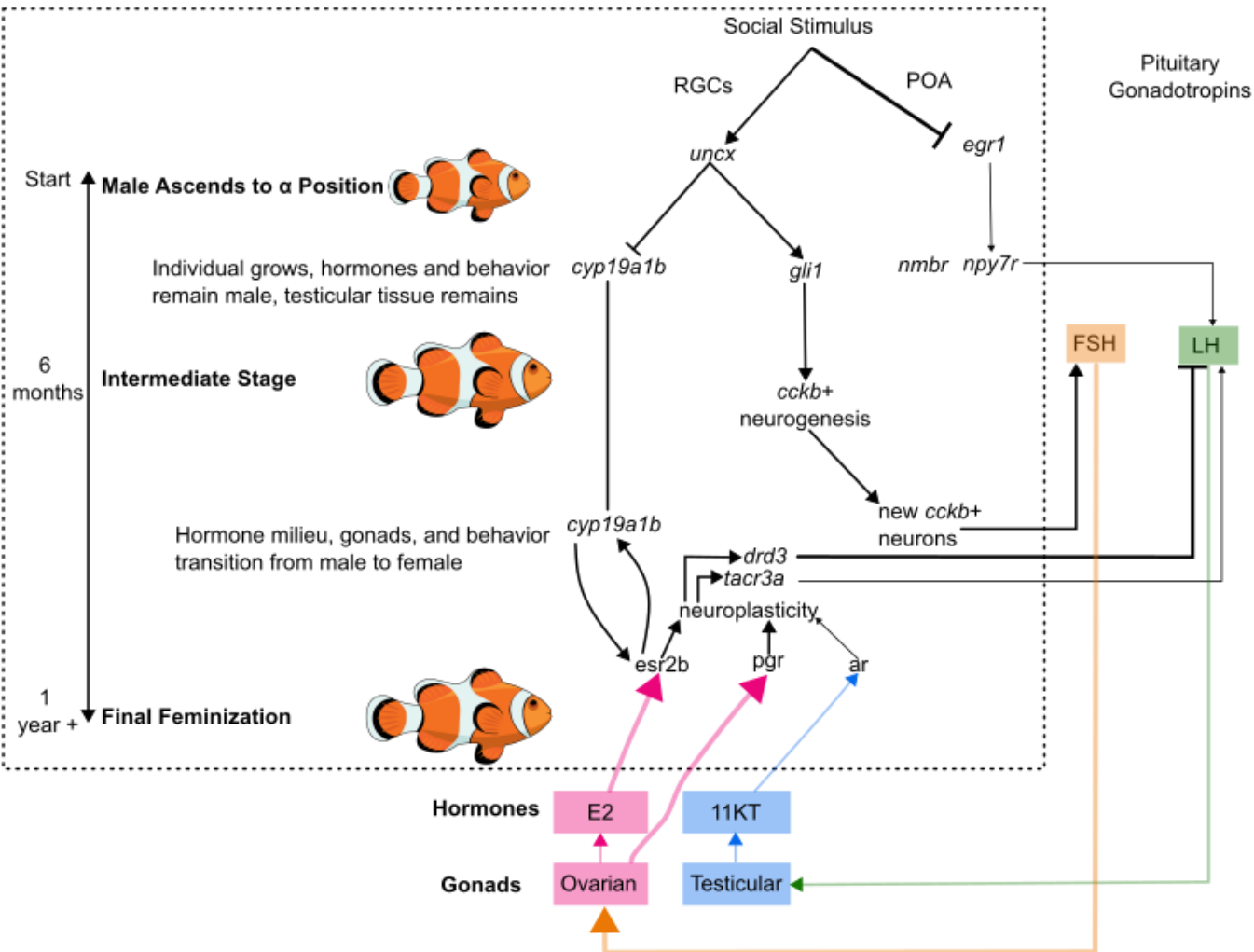
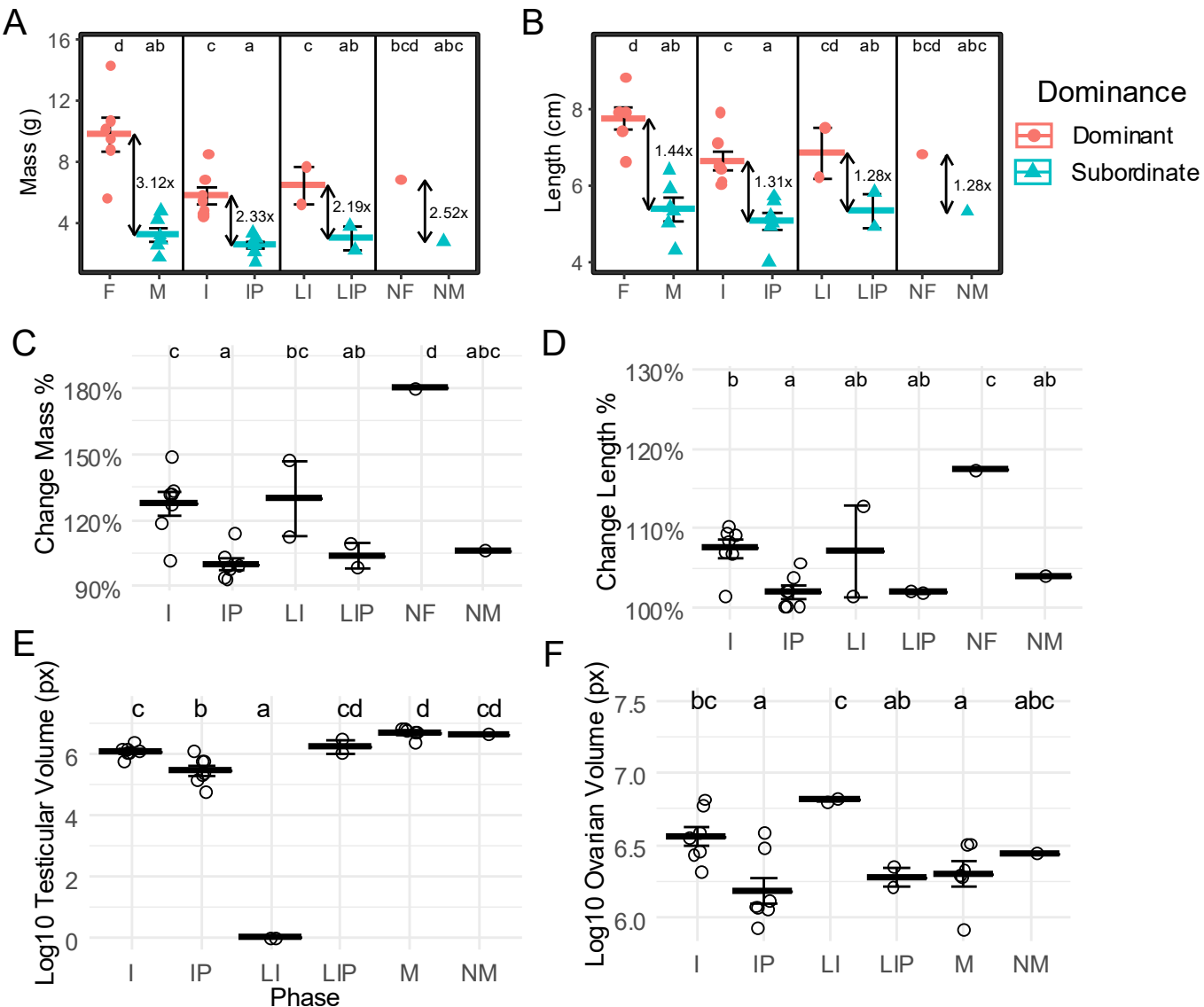


Fig.7 Schematic outlining our model of the changes that occur during protandrous sex change in *A. ocellaris*. Thicker lines indicate increased output, thinner lines indicate decreased output. Arrows indicate stimulation and blunt ends indicate inhibition.



Supplementary Fig. 1.

(A) Body mass by phase.

(B) Body length by phase. Plots are subdivided such that the dominant and subordinate individuals from each pair appear together reflected by shape and color.

(C) Change in body mass by phase.

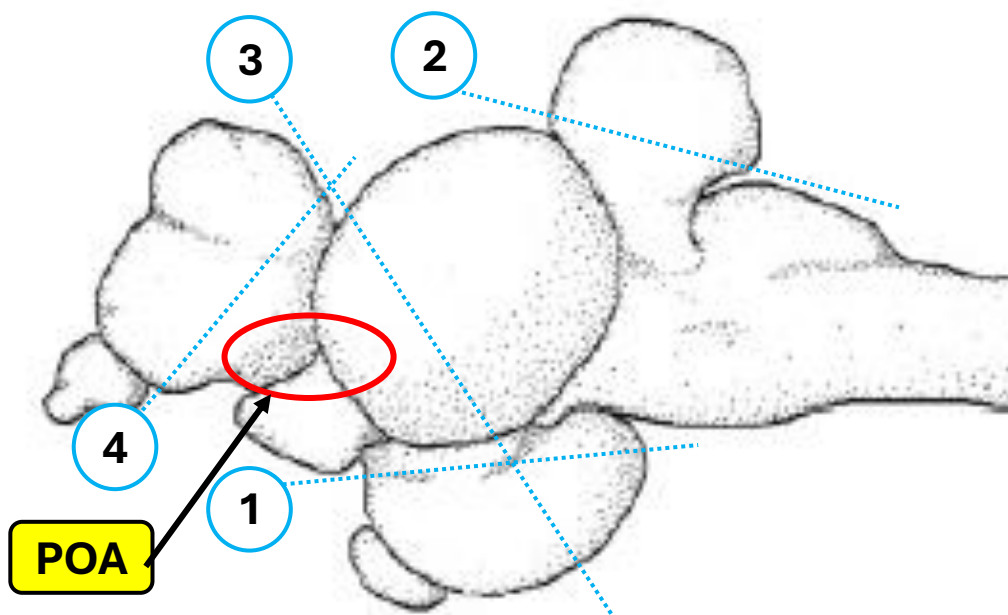
(D) Change in length by phase.

(E) Log10 testicular volume estimate by phase.

(F) Log10 ovarian volume estimate by phase.

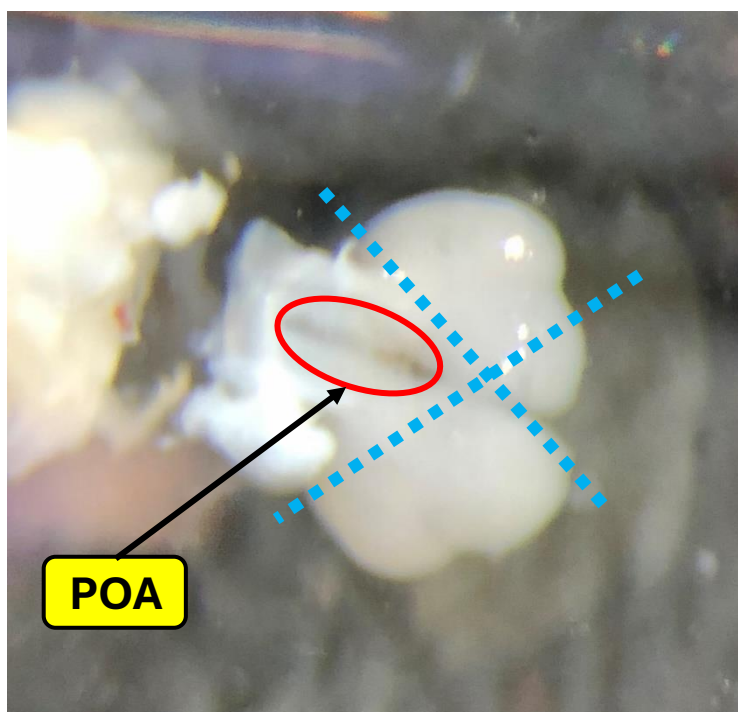
Crossbar indicates mean, error bars indicate standard error of the mean (SEM). Letters indicate significance: groups that do not share a letter are significantly different. In A, B Arrows indicate fold-difference between the two groups.

A



Blue dotted lines represent cuts made

B



Supplementary Fig.2.

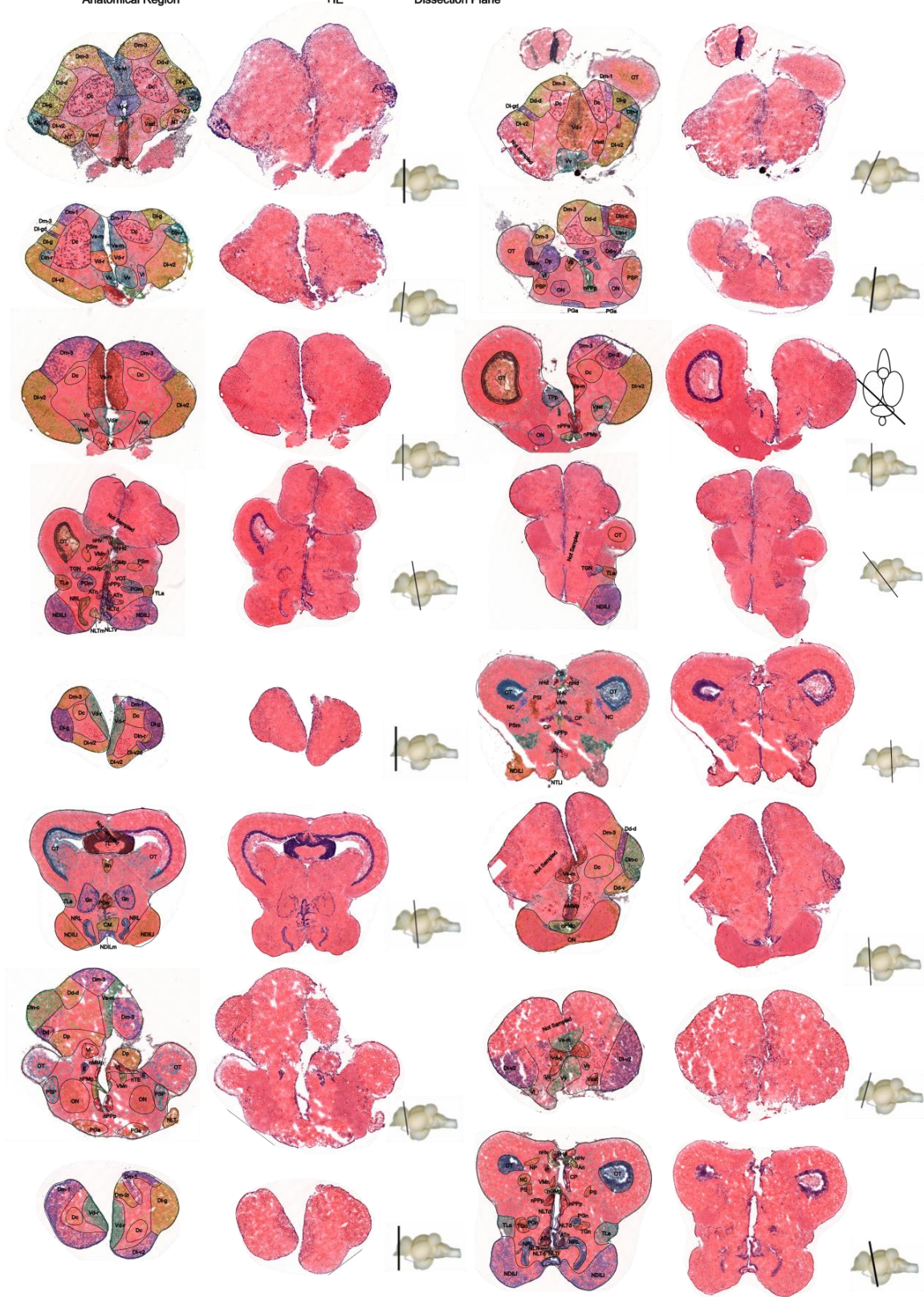
(A) Side view of anemonefish brain with dissections indicated by dashed blue lines. The number indicates the order of cuts.

(B) Top down view of anemonefish brain with dissections indicated by dashed blue lines. In A and B, the POA is highlighted with a red circle

Anatomical Region

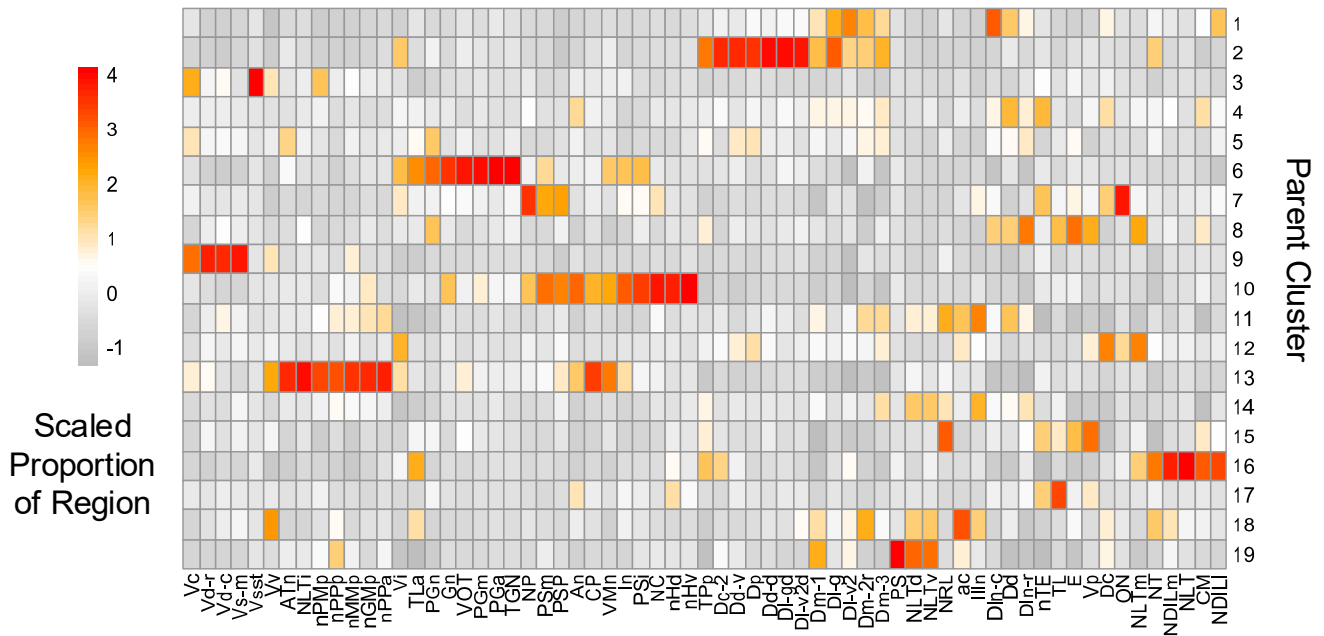
HE

Dissection Plane

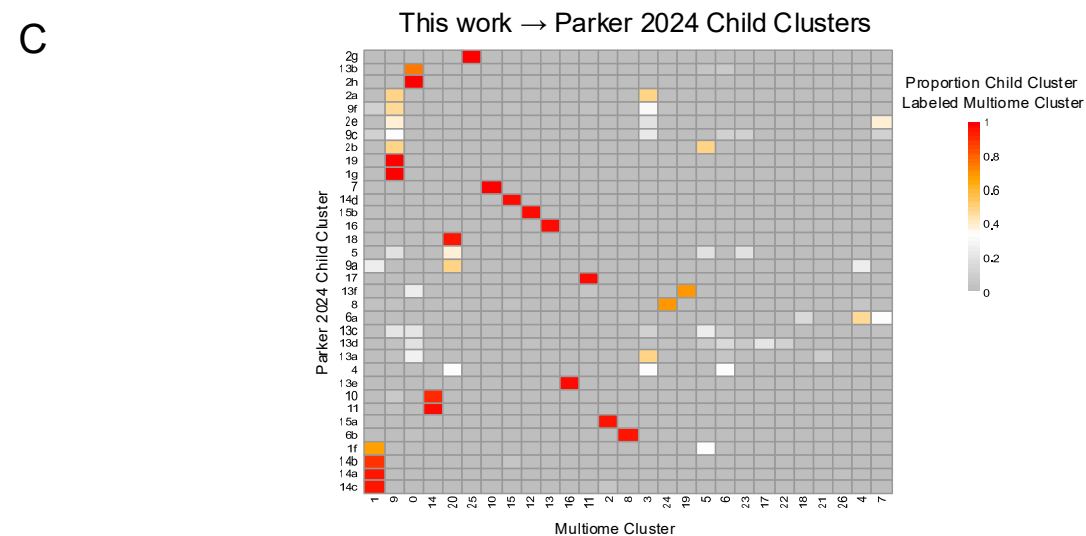
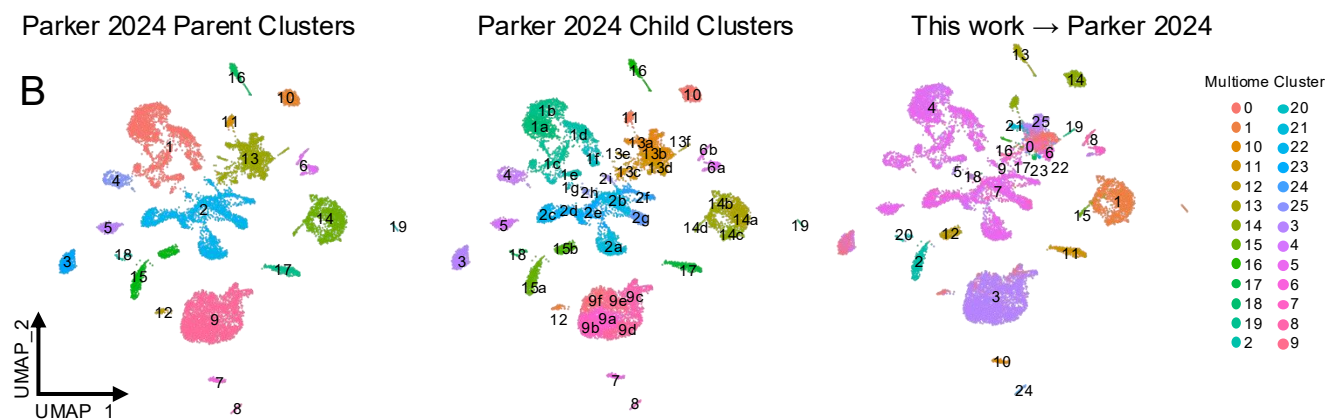
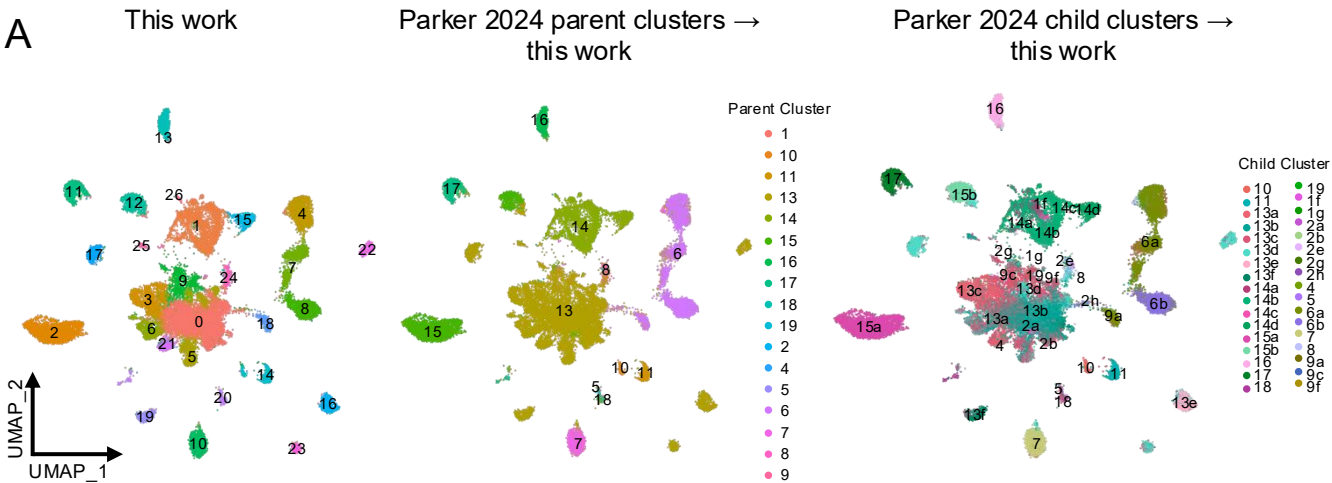


Supplementary Fig.3. Drawings of each section sequenced with Visium HD with neuroanatomical regions labeled. The middle column displays the unlabeled hematoxylin and eosin stain and the right column shows a small brain from the side with a black line indicating the sectioning plane for the given section.

Parker 2024 Parent Clusters

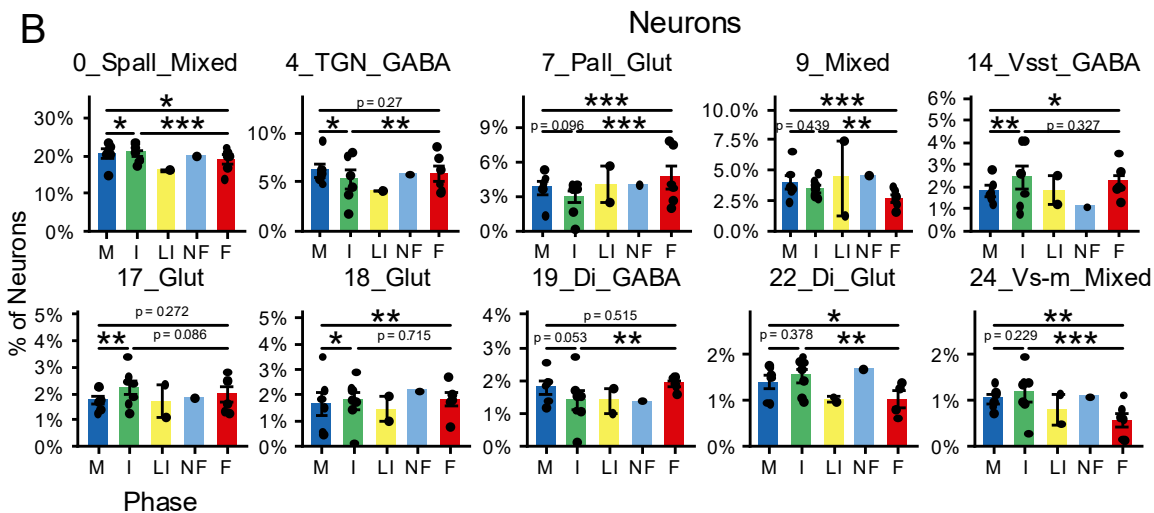
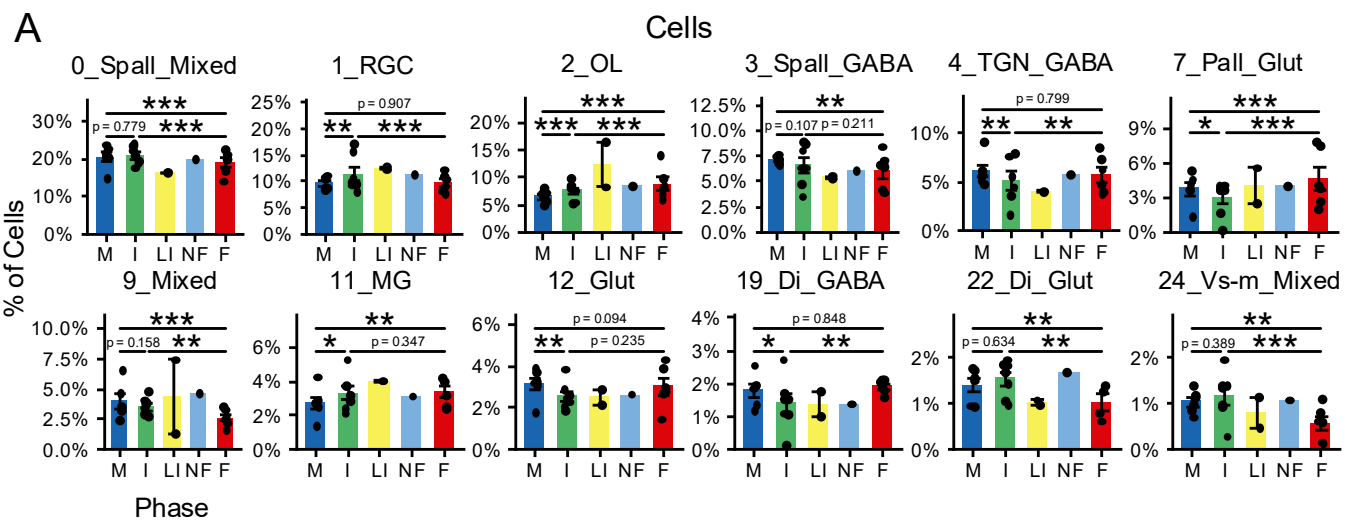


Supplementary Fig.4 Heatmap of Parker 2024 parent clusters mapped onto the Visium HD data in this work. The color in the heatmap indicates the scaled proportion of cells from each parent cluster composing each neuroanatomical region.



Supplementary Fig.5 Results of label transfer from Parker 2024 dataset.

- (A) Projection of Parker 2024 parent and child clusters onto the current work.
- (B) Clusters from the current work projected onto the Parker 2024 object.
- (C) Heatmap showing the proportion of each child cluster in the Parker 2024 labeled with the clusters from this work in a multiome to Parker 2024 child cluster mapping.

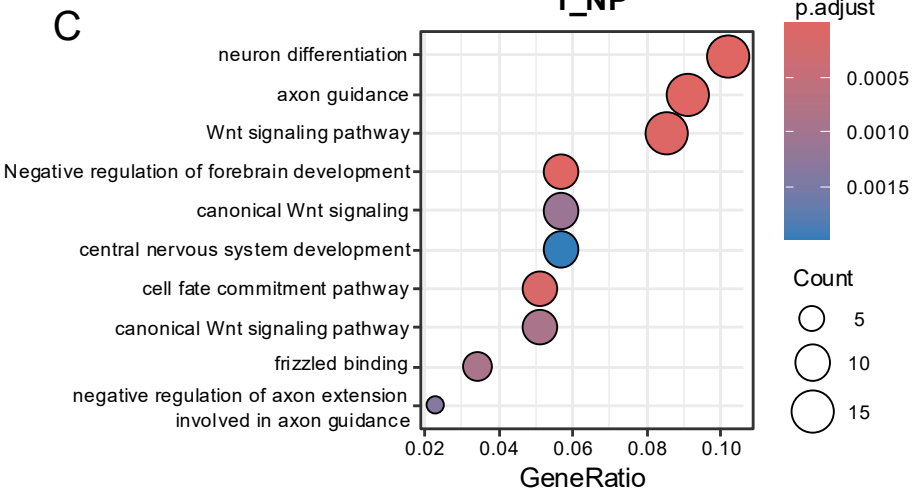
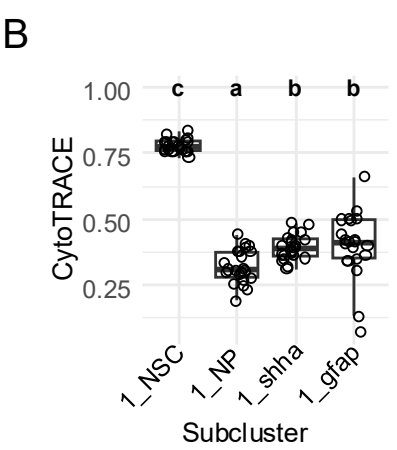
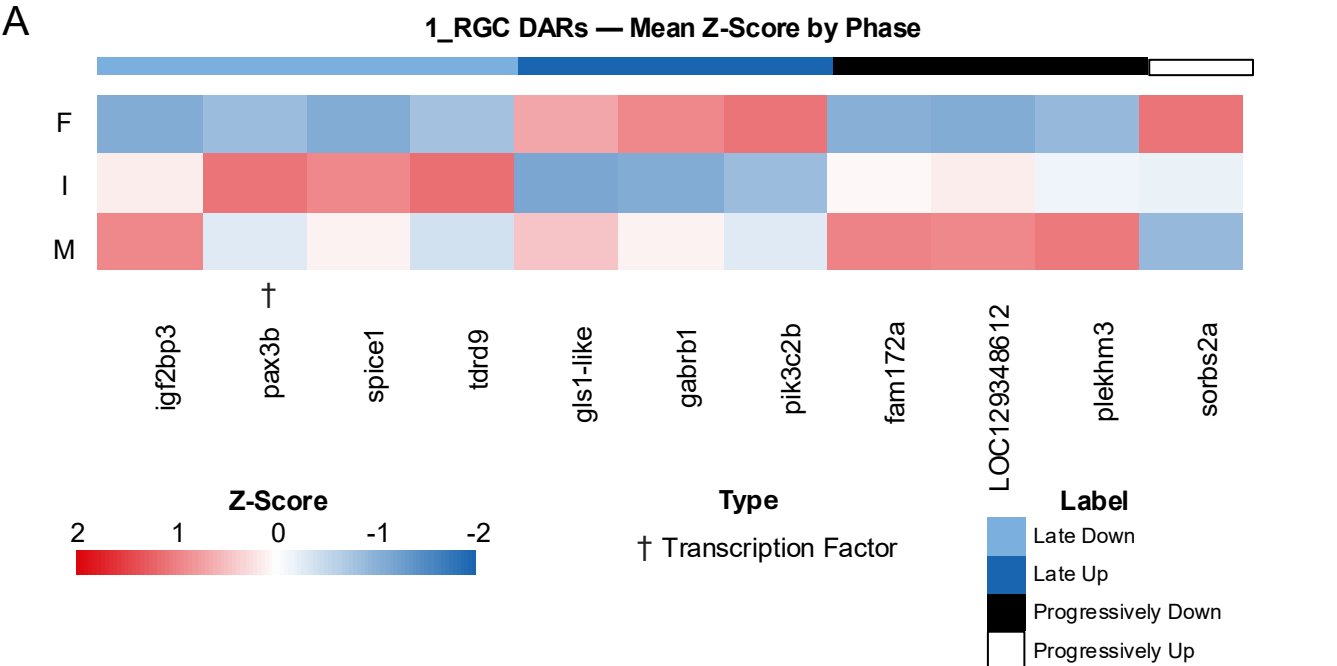


Supplementary Fig.6.

(A) Percent of all cells belonging to each multiome cluster.

(B) Percent of all neurons belonging to each multiome cluster.

Only significant results are shown. Points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.



Supplementary Fig.7.

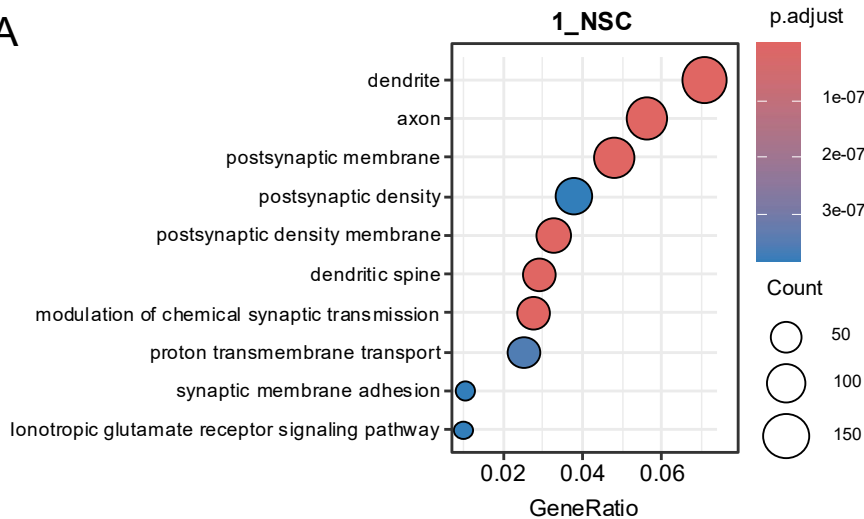
(A) DARs in 1_RGC. Cells are colored by z-score of the mean accessibility or expression for each phase. The top bar indicates the classification for how each gene or region changes sex, and the symbols below the matrix indicate what kind of gene is differentially accessible or differentially expressed.

(B) CytoTRACE score by 1_RGC subcluster. Letters above indicate significance, if two subclusters share a letter they are not significantly different.

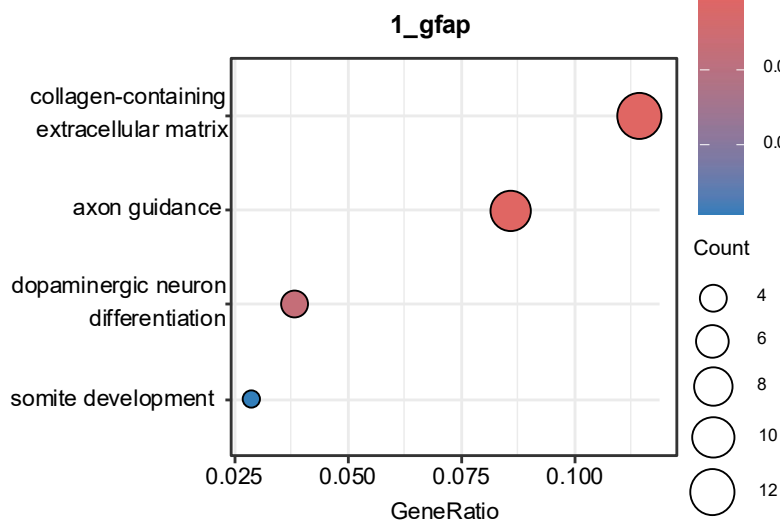
(C) Gene ontology analysis of marker genes for 1_NP.

Color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and "GeneRatio" indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list.

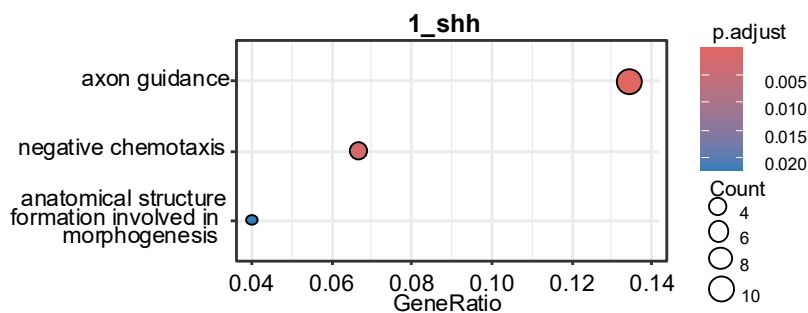
A



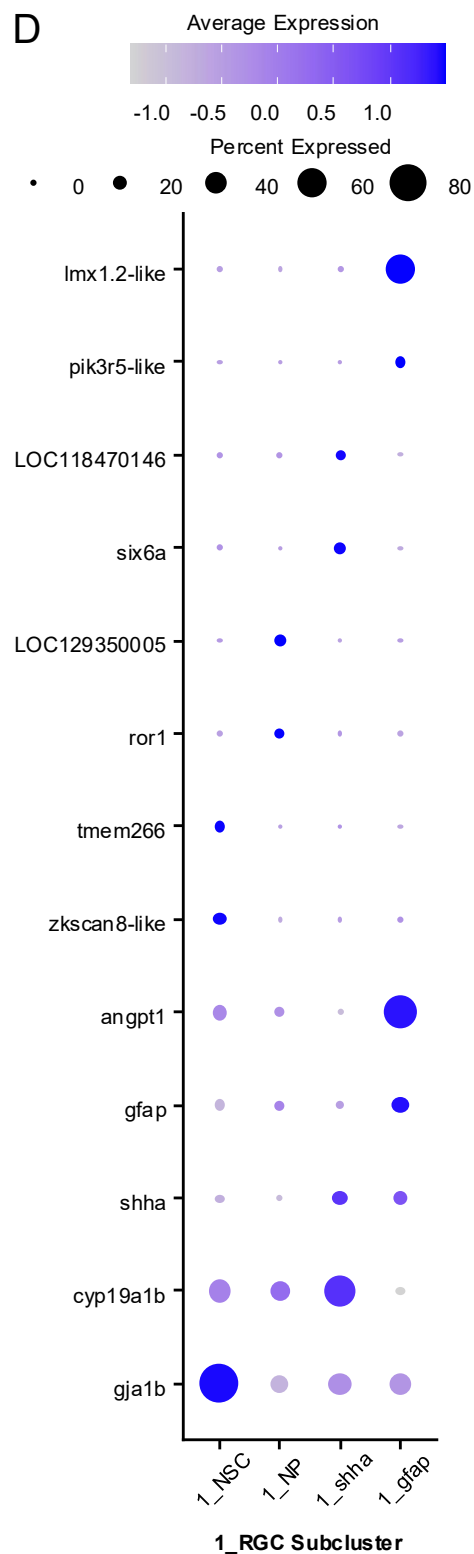
B



C



D



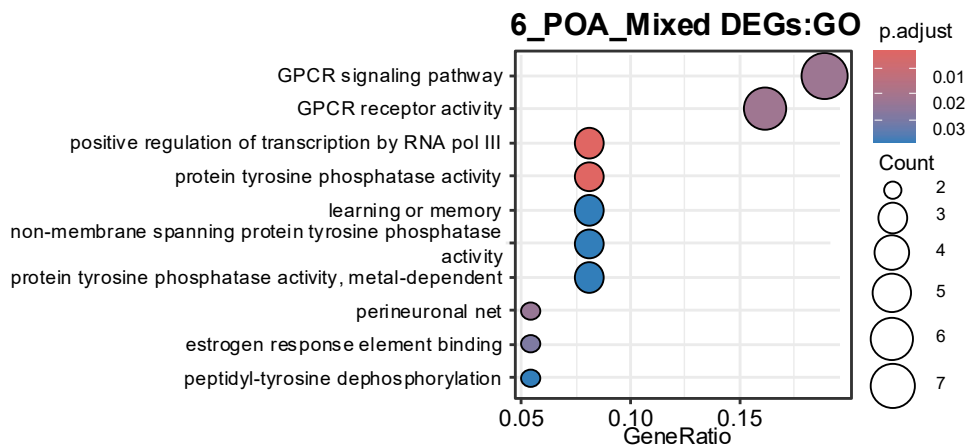
Supplementary Fig.8. Gene ontology (GO) analysis for the marker genes of

(A) 1_NSC,

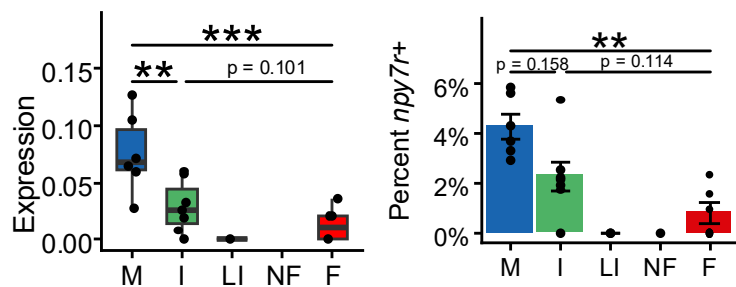
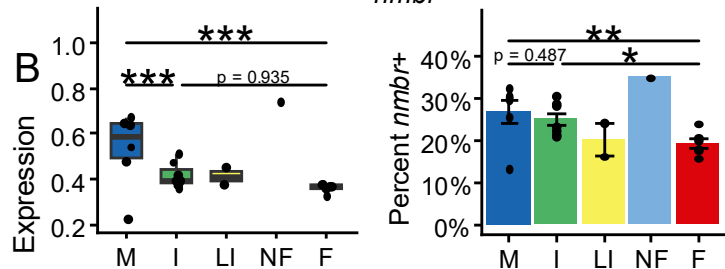
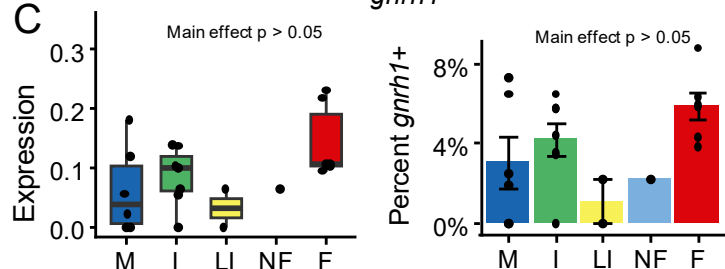
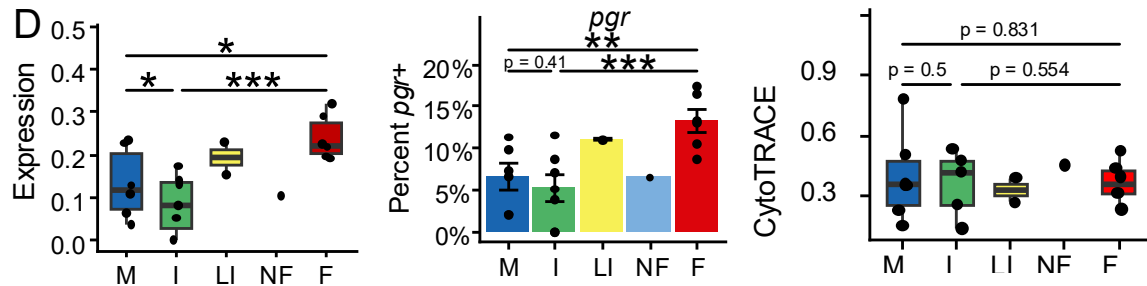
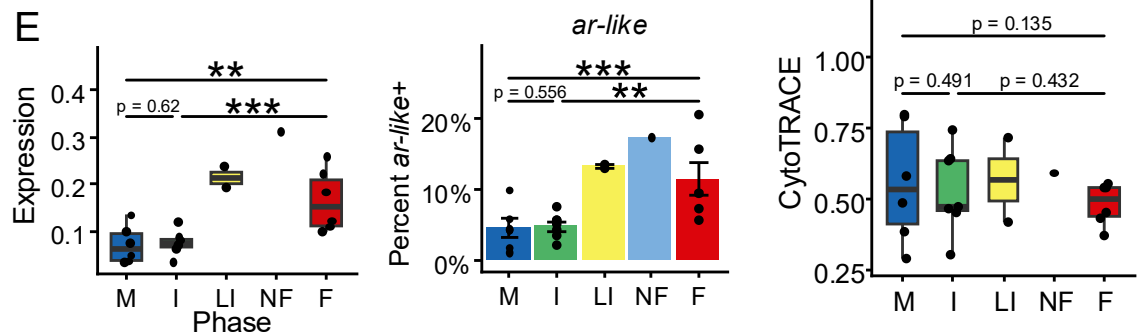
(B) 1_gfap, and

(C) 1_shha.

In A-C, color indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list. (D) Canonical and data-driven marker genes for each 1_RGC subcluster

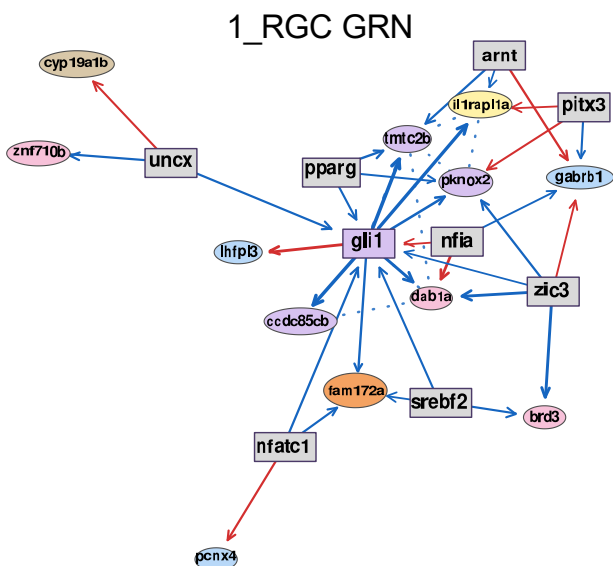


Supplementary Fig.9. GO analysis of all of the DEGs in 6_POA_Mixed. Ccolor indicates false discovery rate adjusted p value, point size indicates the number of genes annotated with that term, and “GeneRatio” indicates the proportion of genes annotated with that GO term divided by all annotated genes in our list.

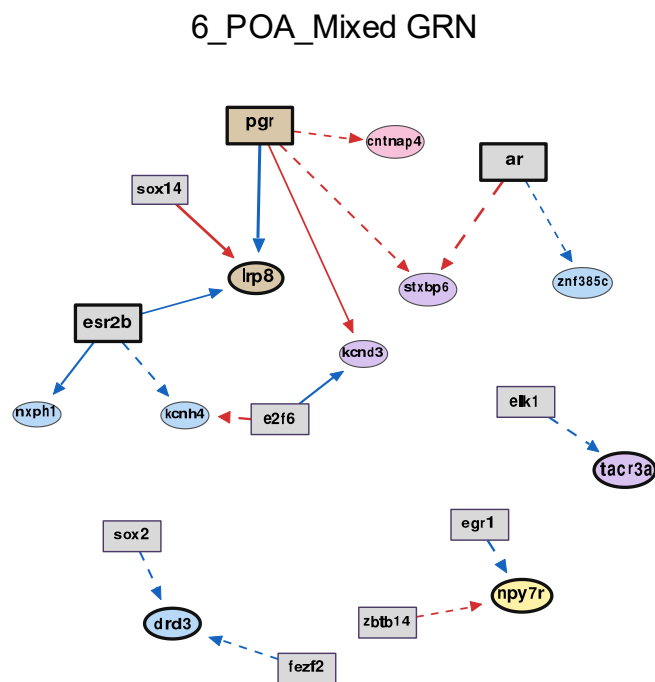
A*npy7r***B***nmb***C***gnrh1***D****E**

Supplementary Fig.10. Expression (left) and percent of 6_POA_Mixed (right) expressing
(A) *npv7r*,
(B) *nmbr*,
(C) *gnrh1*. Expression (left) and percent of 6_POA_Mixed (middle), and CytoTRACE score
(right) of cells expressing
(D) *pgr*,
(E), *ar-like*.
Points indicate the mean for each individual. Asterisks: “*”: $p < 0.05$, “**”: $p < 0.01$, “***”: $p < 0.001$.

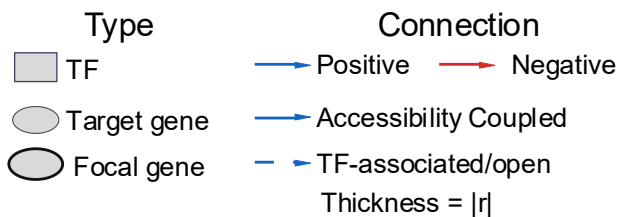
A



B



Classification



Supplementary Fig. 11 xx Justin please write